



RESEARCH METHODS- II

Prerequisite: Research Methodology-I

Course Code: ECON-318



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Part-I

Unit 01: Measurement: Scaling, Reliability and Validity:

INTRODUCTION:

Measurement in research refers to the process of assigning numbers or labels to variables, objects, or events according to specific rules. It helps in gathering data to analyze and draw meaningful conclusions. Three important aspects of measurement are **scaling**, **reliability**, and **validity**.

- **Scaling** is the method of assigning numbers to represent characteristics or qualities of objects. For example, rating a movie on a scale from 1 to 5 stars is a type of scaling.
- **Reliability** refers to the consistency of a measurement. If a scale or tool gives the same results repeatedly under the same conditions, it is reliable. For instance, a weighing machine showing the same weight for a person multiple times reflects reliability.
- **Validity** ensures that the measurement accurately captures what it is intended to measure. For example, a math test should measure math skills, not language ability.

Example of Measurement: Scaling, Reliability, and Validity

Imagine you are conducting a survey to find out how much people like a new chocolate brand. Here's how **scaling**, **reliability**, and **validity** work:

1. **Scaling:** You ask people to rate the chocolate on a scale of 1 to 5, where:
 - 1 means "I don't like it at all"
 - 5 means "I love it!"

This scale helps you measure their feelings in numbers.

2. **Reliability:** If the same person tastes the chocolate again tomorrow and gives the same rating, your scale is reliable. Consistency in their response shows reliability.
3. **Validity:** Your question should actually measure how much they like the chocolate, not something else. For example, if you ask, "Do you like sweets?" instead of asking about the specific chocolate, the results may not be valid for your study.

FOUR TYPES OF MEASUREMENT SCALES AND THEIR PROPERTIES:

In research, **measurement scales** are ways of assigning numbers or labels to data. There are four types, each with specific properties and uses:

1. Nominal Scale: The Nominal Scale is the simplest type of measurement scale, where data is categorized or labeled without any specific order or ranking. The main property of this scale is that the categories are unique and do not overlap.

Additionally, there is no logical order between these categories. For example, types of fruits such as apple, banana, and orange, or categories like gender (male, female, other), are nominal data. These categories simply serve as names or labels to identify and classify different groups.

2. Ordinal Scale: In the Ordinal Scale, data is categorized, but the categories have a logical order or ranking. However, the differences between the ranks are not equal or measurable. The main properties of this scale are that the categories can be ranked, but the exact differences between them cannot be determined. For example, rating a movie as 1 (Bad), 2 (Average), 3 (Good), or 4 (Excellent) shows a ranking, but the gap between these ratings may vary. Similarly, education levels such as High School, Bachelor's, Master's, and PhD follow a logical order, but the difference in time or effort between these levels is not uniform.

3. Interval Scale: The Interval Scale measures data on a scale where the differences between values are meaningful and consistent, but there is no true zero point. This

means that zero on the scale does not represent the complete absence of what is being measured. The key properties of this scale are that the intervals between values are equal, but zero does not signify "nothing." For example, in temperature measured in Celsius or Fahrenheit, the difference between 10°C and 20°C is the same as the difference between 20°C and 30°C. However, 0°C does not mean there is no temperature; it is simply a reference point on the scale.

4. The Ratio Scale: The Ratio Scale is the most advanced type of measurement scale. It includes all the properties of the interval scale but also has a true zero point, where zero represents "nothing." The key properties of this scale are that it has equal intervals between values, and the presence of a true zero allows for meaningful comparisons, such as ratios. For example, in weight measurement, 0 kg means no weight at all, and 10 kg is exactly double 5 kg. Similarly, in measuring distance, 0 meters means no distance, and 20 meters is twice as far as 10 meters. This scale allows for the most precise and meaningful analysis.

Summary Table:

Scale	Order	Equal Differences	True Zero	Example
Nominal	No	No	No	Fruits, Gender
Ordinal	Yes	No	No	Movie Ratings, Education Level
Interval	Yes	Yes	No	Temperature (Celsius, F)
Ratio	Yes	Yes	Yes	Weight, Distance

These scales help researchers choose the right way to measure and analyze data!

RATING SCALES: DEFINITION AND ITS TYPES:

A **rating scale** is a tool used in surveys, questionnaires, and research to measure people's opinions, attitudes, or behaviors. It allows respondents to evaluate

or rate something by assigning it a value, typically using numbers, symbols, or descriptive labels. For example, rating your experience in a restaurant as **1 (Very Bad)** to **5 (Excellent)** is a type of rating scale.

Types of Rating Scales:

1. Likert Scale: It measures the level of agreement or disagreement with a statement. Example: "I enjoy using this product." Strongly Disagree (1), Disagree (2), Neutral (3), Agree (4), Strongly Agree (5).

Use: Common in attitude surveys and social science research.

2. Numerical Rating Scale (NRS): A scale that uses numbers to rate the intensity or preference. Example: "Rate your pain on a scale of 1 to 10." 1 (No pain) to 10 (Worst pain imaginable).

Use: Often used in healthcare and feedback forms.

3. Graphic Rating Scale: A visual scale where respondents mark their rating on a line or a graphic. Example: A line from "Very Dissatisfied" to "Very Satisfied," where you place a mark to indicate your level of satisfaction.

Use: Used for customer satisfaction or emotional responses.

4. Semantic Differential Scale: Measures attitudes by asking respondents to rate something on a scale between two opposite adjectives. Example: "How would you describe the service?" Poor --Excellent, Slow --Fast

Use: Used in product evaluations and marketing research.

5. Checklist Rating Scale: Provides a list of items or behaviors that respondents can check off as applicable. Example: "Check all that apply to describe your experience:

Friendly staff

Quick service

Clean environment."

Use: Useful in behavioral assessments or feedback forms.

Summary Table

Type	How It Works	Example
Likert Scale	Measures agreement/disagreement with statements	Strongly Agree to Strongly Disagree
Numerical Rating Scale	Rates intensity or preference using numbers	Pain level 1 to 10
Graphic Rating Scale	Uses visuals or lines to represent ratings	Line from "Dissatisfied" to "Satisfied"
Semantic Differential Scale	Rates between two opposite words	Poor ----- Excellent
Checklist Rating Scale	List of options to check off	Friendly staff, Quick service

These rating scales make data collection easier and more structured, helping researchers analyze people's thoughts or experiences effectively.

RANKING SCALES: DEFINITION AND ITS TYPES. EXAMPLES AND EXERCISES TO PRACTICAL SCALES.

Ranking Scales: Ranking scales are tools used to measure preferences or priorities by asking people to arrange options in a specific order, usually from most preferred to least preferred. These scales are helpful in identifying preferences, priorities, or importance levels in decision-making.

1. Simple Ranking Scale: In a simple ranking scale, respondents are asked to rank a set of items or options in order of preference or importance. This type of scale does not allow for ties, so each option must have a unique rank.

Example: Rank the following fruits in order of preference (1 = Most Preferred, 4 = Least Preferred):

1. Apple
2. Banana

3. Orange

4. Mango

- **Practical Exercise:** Ask a group of friends to rank their favorite vacation destinations. Compare and analyze the results.

2. Paired Comparison Scale: In this scale, respondents are presented with two items at a time and asked to choose their preferred option. This process is repeated for all possible pairs, making it easier to compare a large number of items.

- **Example:** Which do you prefer?
 - Apple vs. Banana
 - Mango vs. Orange
 - Apple vs. Orange
- **Practical Exercise:** Compare preferences for a product by creating pairs of features (e.g., price vs. quality) and collecting responses.

3. Forced Ranking Scale: This scale requires respondents to rank all items in a list, even if they do not have a clear preference. It forces them to assign priorities, which can reveal hidden preferences.

Example: Rank the following factors based on their importance when choosing a job (1 = Most Important, 5 = Least Important):

- Salary
- Work-life balance
- Career growth
- Location
- Company reputation
- **Practical Exercise:** Have students rank the importance of study habits like time management, group study, and self-study. Discuss the outcomes.

4. Rank Order Distribution Scale: This type of scale allows respondents to distribute ranks across a set of items, often with a fixed total (e.g., 10 points). Respondents can assign the same rank to multiple items, making it flexible.

Example: Allocate 10 points across these qualities based on their importance in a leader:

- Honesty
- Communication skills
- Vision
- Decision-making
- **Practical Exercise:** Conduct a class activity where students allocate points to various extracurricular activities to show their interest levels.

Practical Importance

Ranking scales provide clear insights into people's priorities and preferences. Exercises like ranking foods, hobbies, or career factors make this concept easy to apply and understand in real life.

CONSTRUCT VALIDITY: INTERNAL AND EXTERNAL.

Construct validity is about ensuring that a test or measurement tool truly measures the concept or idea it is supposed to measure. For example, if you are creating a test to measure "intelligence," the test should actually measure intelligence, not something else like memory or motivation. Construct validity ensures the results are meaningful and accurate for the intended concept.

Construct validity is divided into two main types: **internal validity** and **external validity**.

1. Internal Validity: Internal validity is about whether the study or test measures the concept it claims to measure, without being affected by other factors. It checks if the results are reliable and accurate **within the study itself**. High internal validity means the outcomes of the test are directly related to the concept being studied.

Example: If a study claims that a new teaching method improves student learning, internal validity ensures that it is the teaching method, not other factors

(like extra homework or better classroom environment), that caused the improvement.

- **How to Improve Internal Validity:**

- Use proper controls to minimize outside influences.
- Randomize participants to avoid bias.
- Ensure clear and consistent procedures during testing.

2. External Validity: External validity is about whether the results of the study or test can be generalized to other situations, people, or settings. It checks if the findings apply **outside the study environment**. High external validity means the results are meaningful in real-world scenarios.

Example: If a survey shows that a specific diet helps people lose weight, external validity ensures that this result applies to people in different age groups, regions, or lifestyles, not just the participants of the study.

- **How to Improve External Validity:**

- Use a diverse group of participants.
- Conduct the study in real-world settings.
- Repeat the study in different environments to confirm the findings.

GOODNESS OF MEASURES, RELIABILITY AND STABILITY MEASURES

Goodness of measures refers to how well a test, tool, or method measures what it is supposed to measure. In research, it is important to ensure that the measurement process is reliable, stable, and valid so that the results are accurate and meaningful. Two key aspects of goodness of measures are **reliability** and **stability**.

1. Reliability: Reliability is the consistency of a measurement tool. A measure is considered reliable if it gives the same results when repeated under similar conditions. It ensures that the results are not random or influenced by external factors.

Types of Reliability:

1. **Test-Retest Reliability:** Measures consistency over time. For example, if a personality test is given to the same group of people twice, the results should be similar.
2. **Inter-Rater Reliability:** Measures consistency between different observers or raters. For example, if two teachers grade the same essay, their evaluations should align.
3. **Internal Consistency:** Ensures that all parts of a test measure the same concept. For example, in a satisfaction survey, all questions about satisfaction should produce consistent results.

Improving Reliability:

- Use standardized procedures during testing.
- Train observers or raters to ensure consistent scoring.
- Increase the number of items in a test to cover the concept thoroughly.

2. Stability: Stability is a type of reliability that focuses on whether a measurement tool produces consistent results over time. If a measure is stable, it means that results will not change significantly when the test is repeated after some time, assuming the conditions remain the same.

Example of Stability: Imagine weighing an object on the same scale multiple times over a week. If the weight remains constant, the scale is stable. Similarly, if a survey about job satisfaction produces similar results when conducted twice with the same employees, it shows stability.

Improving Stability:

- Use tools and tests with proven reliability.
- Minimize external factors that can affect results (e.g., environmental changes).
- Repeat tests over time to verify consistency.

Practical Importance

Ensuring the **goodness of measures** by focusing on reliability and stability helps researchers trust the data they collect. For example:

- In medical research, reliable tools ensure accurate diagnosis.
- In education, stable tests help measure student progress consistently.

By designing reliable and stable measures, researchers can gather meaningful data and make sound conclusions.



Unit No.02: Data Collection Methods

What is Data in Research?

Data in research refers to raw, unprocessed facts and figures collected during a study. It is the foundation of research, providing the necessary inputs for analysis, interpretation, and decision-making. Data can be quantitative (numbers, measurements) or qualitative (descriptions, feelings, observations). Data alone does not provide meaning; it needs to be analyzed and interpreted to derive insights.

What is Information in Research?

Information in research is the processed, organized, and meaningful interpretation of data. When raw data is analyzed, patterns, trends, and conclusions emerge, turning data into information. Information is used to make decisions, develop theories, or validate hypotheses in research. Information provides context and understanding, helping researchers communicate findings effectively.

Example: Data: 85, 90, 78 (test scores). Information: The average test score is 84, indicating good performance in the class.

Key Difference Between Data and Information:

- **Data:** Raw and unorganized facts. Example: A list of numbers or observations.
- **Information:** Processed and meaningful insights derived from data. Example: A report showing trends based on the numbers.

SOURCES OF DATA:

In research, **sources of data** refer to the origin or method through which information is collected. These sources are broadly classified into **primary sources** and **secondary sources**.

1. Primary Sources of Data: Primary data is collected directly from the original source by the researcher for the specific purpose of the study. It is first-hand, raw data that has not been processed or interpreted by anyone else.

Examples of Primary Data Sources:

- I. **Surveys:** Collecting data through questionnaires or interviews. Example: Asking students about their study habits.
- II. **Experiments:** Conducting controlled experiments to observe outcomes. Example: Testing a new drug to measure its effectiveness.
- III. **Observations:** Watching and recording behaviors or events. Example: Observing customer behavior in a store.
- IV. **Interviews:** Direct conversations with individuals or groups. Example: Asking employees about job satisfaction.
- V. **Focus Groups:** Group discussions to gather insights. Example: Discussing consumer preferences for a product.

2. Secondary Sources of Data: Secondary data is collected and processed by someone else and made available for use. It is often analyzed data from previous studies, reports, or published documents.

Examples of Secondary Data Sources:

1. **Published Research:** Articles, journals, or academic papers. Example: A study on global warming trends.
2. **Government Reports:** Official statistics or publications. Example: Census data or employment statistics.
3. **Books and Reference Materials:** Information available in libraries or online. Example: A textbook on economic theories.
4. **Websites and Databases:** Online platforms providing data. Example: World Bank or United Nations reports.
5. **Historical Records:** Archival documents, letters, or photographs. Example: Historical weather patterns from old records.

EXPERIMENTS, SURVEY DESIGN, FIELD STUDY:

1. Experiments: An **experiment** is a method of research where researchers manipulate one or more variables (things that can change) to see how they affect

other variables. This is done in a controlled environment to test a hypothesis (an idea that can be tested). Experiments are mainly used to establish cause-and-effect relationships.

Example: Imagine a researcher wants to know if studying in a quiet room improves test scores. They have two groups of students: one group studies in a quiet room, and the other group studies in a noisy room. After the study, the researcher compares the test scores of both groups to see if the quiet room led to better scores.

- **Independent Variable** (manipulated): The study environment (quiet vs noisy).
- **Dependent Variable** (measured): The test scores.

Advantages:

- Can establish cause and effect.
- Researchers can control variables to avoid outside influences.

Disadvantages:

- Experiments can sometimes be artificial, so they may not always reflect real-life situations.

2. Survey Design: A survey is a method used to collect information from a large group of people by asking them questions. Surveys are useful for gathering opinions, attitudes, or behaviors from a population. The design of a survey involves how questions are structured, the format of the survey, and how it will be distributed to participants.

Types of Survey Questions:

1. **Closed-ended questions:** These questions have specific answer choices.
Example: "Do you like online learning?" (Yes/No)
2. **Open-ended questions:** These questions allow respondents to answer freely.
Example: "What do you think about online learning?"

Survey Design Example:

A researcher wants to know how many people prefer working from home. They

design a survey with questions like "Do you prefer working from home?" (Yes/No) and "Why do you prefer working from home?" (Open-ended). They send the survey online to a large number of employees.

Advantages:

- Can gather data from a large number of people quickly.
- Relatively easy to design and administer.

Disadvantages:

- Responses may not always be accurate or truthful.
- Survey design errors (like unclear questions) can affect results.

3. Field Study: A field study is a type of research conducted outside the laboratory, in real-world settings. Researchers observe people, events, or situations as they naturally occur, without manipulating any variables. Field studies are often used to understand behaviors or social phenomena in their natural environment.

Example: A researcher wants to study how children behave in a park. They visit the park and observe children playing, talking, and interacting with each other. They don't interfere with the children's behavior but simply record what they see. This is a qualitative approach where researchers are looking for patterns and understanding behavior in a natural setting.

Advantages:

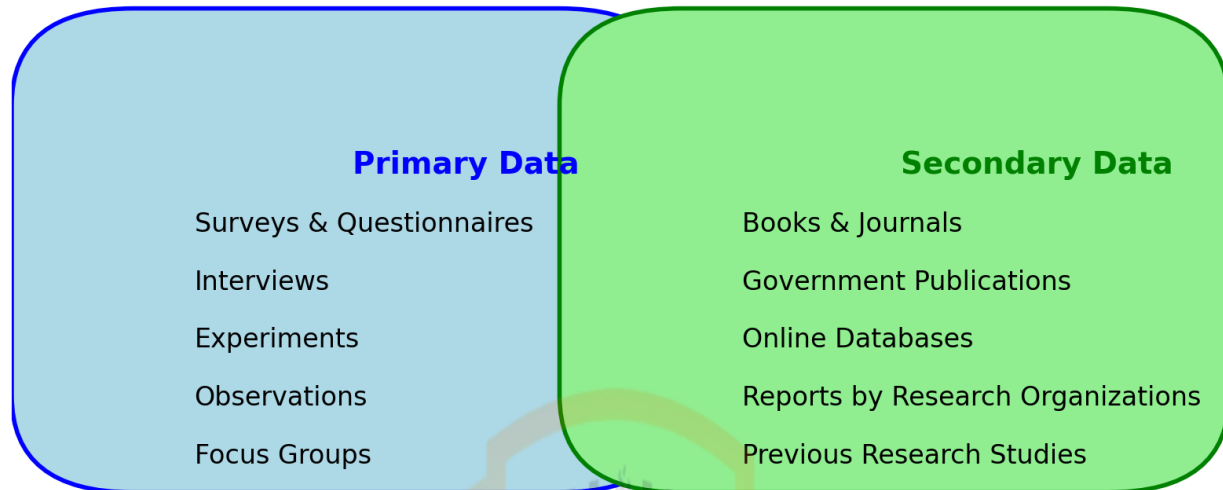
- Provides insights into real-world behaviors and situations.
- More natural and authentic compared to controlled experiments.

Disadvantages:

- Lack of control over the environment, which can make it harder to pinpoint specific causes.
- Data collection can be time-consuming and less structured.

METHODS OF COLLECTING PRIMARY AND SECONDARY DATA:

Classification of Primary and Secondary Data Types



Here is a diagram that classifies the types of primary and secondary data:

- **Primary Data** includes:
 - Surveys & Questionnaires
 - Interviews
 - Experiments
 - Observations
 - Focus Groups
- **Secondary Data** includes:
 - Books & Journals
 - Government Publications
 - Online Databases
 - Reports by Research Organizations
 - Previous Research Studies

This visual representation helps to easily distinguish between the two main categories of data and their respective types.

➤ **Methods of Collecting Primary Data**

Primary data is collected directly from the source for a specific research purpose.

Here are the common methods for collecting primary data:

1. Surveys and Questionnaires: Surveys are used to gather data from a large number of people by asking them structured questions. These questions can be closed-ended (yes/no or multiple choice) or open-ended (free text). **Example:** A researcher conducting a survey on student study habits sends a questionnaire with questions like “How many hours do you study per week?” or “What is your preferred study method?”

Advantages:

- Can collect data from a large group.
- Easy to analyze if the questions are clear.

Disadvantages:

- Respondents might not always be truthful.
- Requires careful question design to avoid bias.

2. Interviews: Interviews involve direct communication between the researcher and the participant. They can be structured (with pre-planned questions) or unstructured (more like a conversation). **Example:** A researcher studying employee satisfaction may conduct one-on-one interviews asking, “What do you enjoy about your job?” and “What improvements would you like to see in the workplace?”

Advantages:

- Provides deeper insights than surveys.
- Allows for follow-up questions and clarification.

Disadvantages:

- Time-consuming.
- Data analysis can be more complex.

3. Experiments: In experiments, researchers manipulate one variable to see how it affects another. This method is used to establish cause-and-effect relationships.

Example: A researcher wants to know if listening to music while studying improves memory retention. They have one group of students study with music and another group study in silence, then compare their test scores.

Advantages:

- Can prove cause-and-effect.
- High level of control over variables.

Disadvantages:

- Artificial setting may not reflect real-world situations.
- Ethical issues may arise in some experiments.

4. Observations: Observation involves the researcher watching and recording the behavior of individuals or groups in their natural environment without interfering. Example: A researcher may observe customer behavior in a store to understand purchasing patterns, like how often they check a product's price before buying it.

Advantages:

- Provides natural, real-world data.
- Useful when direct communication is not possible.

Disadvantages:

- Observer bias can affect results.
- Data collection can be time-consuming.

5. Focus Groups: Focus groups are group discussions led by a moderator. Participants are asked questions about a specific topic, and their responses are used to gather insights. Example: A company may hold a focus group to understand consumer opinions on a new product, asking participants what they like and dislike about it.

Advantages:

- In-depth group insights.
- Participants can stimulate each other's thoughts.

Disadvantages:

- Can be influenced by dominant voices in the group.
- May not represent the larger population.

➤ **Methods of Collecting Secondary Data**

Secondary data is information that has already been collected by someone else for another purpose. It is often found in published sources. Here are common methods of collecting secondary data:

1. Books and Journals: Books and academic journals contain research studies, articles, and theoretical frameworks that researchers can use. Example: A researcher studying economic growth may refer to published books and journals on the topic for background information.

Advantages:

- Provides established and reliable information.
- Saves time and resources.

Disadvantages:

- May not be fully relevant or up-to-date.
- The researcher has no control over the data collection process.

2. Government Publications: Governments publish a variety of reports, statistics, and documents, often available online, covering topics such as population, economy, education, health, and more. Example: Census data or annual reports on unemployment published by the government.

Advantages:

- Reliable and authoritative data.
- Covers large-scale data such as national statistics.

Disadvantages:

- May be outdated or not directly relevant to specific research.
- Sometimes hard to interpret.

3. Online Databases and Repositories: There are many online platforms that offer access to academic papers, research reports, and datasets from various fields. Example: A researcher can access databases like JSTOR, Google Scholar, or government databases to find secondary data related to climate change.

Advantages:

- Easy and quick access to a wide range of data.
- Often free or available through institutions.

Disadvantages:

- Some databases require subscriptions or institutional access.
- Quality control may vary.

4. Reports and Publications by Research Organizations: Research organizations like NGOs, think tanks, or private research firms publish reports on various subjects. These reports may contain findings from studies, market analysis, or policy recommendations. Example: A market research company may publish a report on consumer behavior in a specific industry.

Advantages:

- Can provide specialized data not available elsewhere.
- Often based on extensive research.

Disadvantages:

- May not be impartial (bias from the organization).
- May not be specific to the researcher's needs.

5. Previous Research Studies (Literature Review) : Researchers often use data from previous studies conducted by others. These studies are typically available in academic articles, theses, or research papers. Example: A researcher studying the impact of social media on mental health can refer to previous studies on the same topic for secondary data.

Advantages:

- Saves time in data collection.
- Allows comparison with other studies.

Disadvantages:

- Data may not be fully relevant or accurate for current needs.
- Data collection methods of the original study may differ.

Summary

- **Primary Data:** Directly collected for the research, using methods like surveys, interviews, experiments, observations, and focus groups.
- **Secondary Data:** Data already collected by others, such as through books, government reports, online databases, and previous research.

Both primary and secondary data collection methods have their own strengths and weaknesses, and researchers often use a combination of both to get comprehensive insights.

GUIDELINES FOR QUESTIONNAIRE DESIGN:

Designing an effective questionnaire is crucial for collecting accurate, reliable, and valid data. Here are key guidelines for creating a well-structured questionnaire:

1. Define the Objective:

- **Purpose:** Clearly define the goal of the questionnaire. What do you want to learn? What research question are you answering?
- **Target Audience:** Identify the group of people who will answer the questionnaire, keeping in mind their background, literacy levels, and interests.

2. Question Types:

- **Closed-ended questions:** These limit responses to a set of predefined options (e.g., Yes/No, multiple choice, Likert scale). They are easier to analyze but may not capture all nuances.
- **Open-ended questions:** These allow respondents to answer in their own words. While they provide richer data, they are more difficult to analyze.

- **Scaled questions:** These are used to measure attitudes, opinions, or behaviors on a scale (e.g., Strongly Agree to Strongly Disagree).

3. Question Clarity:

- **Simplicity:** Questions should be easy to understand. Avoid jargon, technical terms, or complex phrasing.
- **Specificity:** Be specific in what you're asking to avoid ambiguity.
- **Single-focus:** Each question should address only one issue or concept at a time, avoiding double-barreled questions (e.g., "Do you like both movies and books?").

4. Question Sequence:

- **Logical Flow:** Organize the questions logically, starting with easier or more general questions and progressing to more complex ones.
- **Group Related Questions:** Questions on similar topics should be grouped together to help respondents stay focused.
- **Sensitive Questions:** If you have sensitive topics, place them later in the questionnaire after establishing trust.

5. Response Options:

- **Balanced Choices:** Ensure that response options for closed-ended questions are balanced and cover all potential answers. For example, avoid using "Always" and "Never" without intermediate options like "Sometimes."
- **Mutually Exclusive Options:** Ensure the options do not overlap, making it easy for respondents to choose only one option.
- **Exhaustive Options:** Provide all possible choices. For example, include an "Other" option with space for respondents to fill in if none of the predefined answers fit.

6. Length of the Questionnaire:

- **Conciseness:** Avoid making the questionnaire too long. A long survey can lead to respondent fatigue, reducing the quality of answers.

- **Time Consideration:** Be aware of the time it will take to complete. Ideally, a questionnaire should take no more than 15-20 minutes.

7. Avoid Leading or Biased Questions:

- Questions should be neutral and not guide respondents toward a specific answer. For example, instead of asking, "How much do you love using our product?" ask "How often do you use our product?"

8. Pilot Testing:

- **Pre-test:** Conduct a pilot test with a small group of respondents to identify unclear questions or areas of confusion. Revise the questionnaire based on feedback.
- **Cognitive Testing:** Ask respondents to explain why they chose certain answers to assess if the questions are understood as intended.

9. Instructions and Definitions:

- **Clear Instructions:** Provide clear and simple instructions at the beginning of the questionnaire. Let respondents know how to answer (e.g., "Circle the answer that best fits").
- **Definitions:** If using terms that may be unfamiliar or technical, provide definitions or examples to help respondents understand.

10. Avoid Negative or Double Negatives:

- **Avoid Negative Framing:** Questions like "Do you disagree with the statement that the product is bad?" can confuse respondents. It's better to phrase it positively.
- **Clear Positive/Negative Statements:** If you use a negative frame, ensure the wording is easy to understand. For example, "Do you not like the product?" should be avoided in favor of "Do you dislike the product?"

11. Ensure Anonymity and Confidentiality:

- **Privacy Assurance:** If the questionnaire asks for personal or sensitive information, reassure respondents that their answers will be confidential or anonymous, and explain how their data will be used.

12. Demographic Questions:

- **Position Demographics at the End:** These questions (e.g., age, gender, income) are typically placed at the end of the survey to avoid influencing answers to other questions.
- **Ensure Relevance:** Only ask demographic questions that are relevant to the research.

13. Offer Thank You/Closing Statement:

- **Gratitude:** At the end, thank respondents for their time and effort in completing the questionnaire. If applicable, inform them about the results or next steps.

By following these guidelines, you can design a questionnaire that effectively captures the information needed for your research, while also being clear and respectful to the respondents.

QUANTITATIVE DATA COLLECTION:

Quantitative data refers to data that is expressed in numerical terms, which can be measured and analyzed statistically. It deals with quantities and can be counted or expressed in numbers. This type of data is objective and focuses on the "how much" or "how many" aspect of phenomena. It is often used to quantify attitudes, opinions, behaviors, or characteristics.

Types of Quantitative Data:

1. **Discrete Data:** This data consists of countable values (whole numbers) and usually represents items that can be counted. For example, the number of students in a class, number of cars in a parking lot, etc.
2. **Continuous Data:** This data consists of measurable quantities and can take any value within a range. For example, height, weight, temperature, or time.

Examples of Quantitative Data:

1. **Age:** The number of years a person has lived (e.g., 25 years old).
2. **Height:** The measurement of how tall someone is (e.g., 5 feet 9 inches).
3. **Income:** The amount of money a person earns (e.g., \$50,000 annually).
4. **Temperature:** The degree of warmth or cold (e.g., 22°C).
5. **Test Scores:** The marks scored by a student in an exam (e.g., 85 out of 100).
6. **Population Size:** The number of people in a city or country (e.g., 1 million people).

Quantitative Data Collection:

Quantitative data collection refers to the process of gathering numerical data that can be used to quantify variables and analyze them using statistical methods. This type of data collection is typically structured and objective, focusing on measurable factors.

Methods of Quantitative Data Collection:

1. **Surveys and Questionnaires:** These are commonly used to collect data from a large number of respondents. The questions are often close-ended, providing responses in numerical form. For example, a survey that asks, "How many times do you exercise each week?" with options like 0, 1, 2, 3, etc.
2. **Experiments:** In experimental research, quantitative data is collected through controlled experiments where variables are manipulated, and outcomes are measured numerically. For example, measuring the growth of plants in different light conditions, where the height of the plants is recorded.
3. **Observations:** Observational studies can be used to collect quantitative data by counting the occurrence of specific events or behaviors. For instance, counting the number of people entering a store during a particular time frame.
4. **Tests/Assessments:** Standardized tests or assessments can be used to collect quantitative data, such as test scores, performance scores, or other evaluations.

For example, standardized IQ tests measure a person's intelligence with a numerical score.

5. **Data from Devices/Technologies:** Many fields use sensors or other technological devices to collect quantitative data, such as temperature readings from a thermometer, speed from a radar gun, or stock market data from financial systems.

Example of Quantitative Data Collection:

Imagine a research study aimed at measuring the impact of study time on student performance. The researcher might:

1. **Survey Students:** Ask how many hours they study per week.
2. **Collect Test Scores:** Gather students' final exam scores.
3. **Data Collection Method:** The data collected (study hours and exam scores) will be numerical (e.g., 5 hours, 10 hours, 90%, 75%).
4. **Analysis:** Using statistical tools, the researcher can analyze the correlation between study hours and exam scores, allowing for objective conclusions about the impact of study time.

In summary, quantitative data is numerical and is often used in scientific and social research to establish patterns, relationships, and trends through statistical analysis. The collection of this data involves structured methods such as surveys, experiments, and standardized tests.

QUALITATIVE DATA COLLECTION:

Qualitative refers to non-numeric data that describes qualities or characteristics. It focuses on exploring ideas, feelings, experiences, and perceptions rather than measuring or quantifying things. This type of data is rich in detail and can provide a deeper understanding of a subject, particularly in understanding complex human behaviors, emotions, and social phenomena. Examples of qualitative data include:

- **Descriptions of experiences** (e.g., "How did you feel about your first day at work?")
- **Personal opinions** (e.g., "What is your opinion on the current education system?")
- **Narratives** (e.g., stories about a person's journey in starting a new business)
- **Observations** (e.g., noting a group's interaction during a meeting)
- **Interviews or open-ended survey responses** (e.g., "What challenges did you face when moving to a new city?")

Qualitative Data Collection: Qualitative data collection refers to the process of gathering data that provides insights into underlying reasons, motivations, and perspectives. This data is typically descriptive and used to understand complex social or cultural phenomena.

Methods of Qualitative Data Collection:

1. Interviews:

- One-on-one conversations between the researcher and the participant.
- Can be **structured**, **semi-structured**, or **unstructured**, depending on how rigidly the questions are pre-defined.
- Example: A researcher may conduct an interview with employees to understand their job satisfaction and the factors influencing it.

2. Focus Groups:

- A group of participants discuss a specific topic guided by a facilitator.
- This method captures different perspectives and group dynamics.
- Example: A company might conduct a focus group with customers to gather feedback on a new product.

3. Observations:

- The researcher observes people in their natural settings without interference.

- Can be **participant** (researcher is involved in the activity) or **non-participant** (researcher just observes).
- Example: A researcher might observe children in a classroom to understand how they interact with technology.

4. **Case Studies:**

- In-depth exploration of a single case (person, event, or group) to gain detailed understanding.
- Example: A researcher could conduct a case study of a community's response to a natural disaster.

5. **Content Analysis:**

- Analyzing textual or visual material (like books, social media posts, or advertisements) to identify themes, patterns, or meanings.
- Example: A researcher may analyze social media posts to understand public reactions to a political event.

6. **Ethnography:**

- Immersive research where the researcher lives and interacts with the community they are studying.
- Example: A researcher might live in a village to understand the cultural practices and social norms of its people.

Example of Qualitative Data Collection:

Imagine a researcher wants to study the experiences of first-time college students in adapting to university life. The qualitative data collection methods could include:

- **Interviews** with students to explore their personal experiences, challenges, and coping strategies.
- **Focus groups** where students share their opinions and feelings about university support systems.

- **Observations** of students in various campus settings, such as during orientation events or while interacting with peers.
- **Diary entries or written narratives** where students document their day-to-day experiences and emotions.

In all these methods, the data collected would be detailed, descriptive, and focus on understanding the underlying reasons behind students' behaviors and attitudes, rather than quantifying how many students experienced specific challenges.

MIXED DATA COLLECTION TECHNIQUES:

Mixed Data Collection Techniques refer to the use of both **qualitative** and **quantitative** methods in a single research study to gather comprehensive data. By combining these two approaches, researchers can take advantage of the strengths of each, providing a fuller understanding of the research problem. Mixed methods allow for both detailed insights (qualitative) and statistical analysis (quantitative), leading to more robust conclusions.

Types of Mixed Data Collection Techniques

1. Sequential Explanatory Design:

- **Process:** First, quantitative data is collected and analyzed. Then, qualitative data is collected to help explain or interpret the quantitative results. Example: A researcher conducts a survey (quantitative) to determine the level of customer satisfaction with a service, then follows up with interviews (qualitative) to understand why customers feel the way they do.

2. Sequential Exploratory Design:

- **Process:** First, qualitative data is collected and analyzed. The findings from this phase are used to guide the development of quantitative data collection, which is used to test or generalize the qualitative results. Example: A researcher conducts in-depth interviews (qualitative) to explore the factors influencing employee

motivation, then develops a questionnaire (quantitative) based on these findings to survey a larger population of employees.

3. **Concurrent Triangulation Design:**

- **Process:** Both qualitative and quantitative data are collected simultaneously, but the two types of data are analyzed separately. The results are then compared or "triangulated" to draw conclusions. Example: A researcher simultaneously administers a survey (quantitative) to measure academic performance and conducts classroom observations (qualitative) to understand student behavior; then compares the findings to draw conclusions.

4. **Concurrent Embedded Design:**

- **Process:** One data collection method (either qualitative or quantitative) is the primary approach, and the other is embedded within it. This allows for one type of data to support or enrich the other. Example: A researcher conducts a large-scale survey (quantitative) on health behaviors, but includes open-ended questions (qualitative) to explore personal experiences related to health, providing deeper insights into the quantitative findings.

Benefits of Mixed Data Collection Techniques:

1. **Comprehensive Understanding:** By combining the depth of qualitative data with the breadth of quantitative data, researchers gain a fuller, more complete picture of the research problem.
2. **Validation and Triangulation:** Mixed methods help confirm findings through cross-validation. If both qualitative and quantitative methods point to the same conclusion, it increases the validity of the results.
3. **Flexibility:** Researchers can address different types of research questions (descriptive, exploratory, or causal) more effectively by using multiple methods.
4. **Richer Data:** Qualitative data provides context and meaning to the numerical data, while quantitative data adds statistical rigor to the descriptive insights.

Challenges of Mixed Data Collection:

1. **Time and Resource Intensive:** Collecting both qualitative and quantitative data can be more time-consuming and resource-demanding than using just one method.
2. **Complexity in Analysis:** Analyzing both qualitative and quantitative data requires expertise in both techniques, and integrating the findings can be challenging.
3. **Conflicting Results:** In some cases, qualitative and quantitative data may not align, which can be difficult to reconcile. Researchers must carefully consider how to interpret such discrepancies.

Example: A study exploring the impact of remote working on employee productivity might use mixed methods as follows:

- **Quantitative:** A survey is sent to employees asking about productivity levels before and after transitioning to remote work, using scales to measure changes in productivity.
- **Qualitative:** In-depth interviews or focus groups are conducted with a smaller group of employees to explore their personal experiences and challenges with remote work.

By combining both methods, the researcher can obtain both statistical evidence on productivity levels and deeper insights into the factors influencing those productivity changes, leading to more comprehensive findings.

SELECTION OF APPROPRIATE METHOD FOR DATA COLLECTION:

Selection of Appropriate Method for Data Collection involves choosing the most suitable technique based on the research objectives, nature of the data, resources available, and the research environment. It is crucial to align the method with the research question to ensure the accuracy and relevance of the data gathered. For example, if the study aims to explore in-depth personal experiences or opinions, qualitative methods such as interviews or focus groups may be ideal. On the other

hand, if the goal is to measure or quantify a specific variable or phenomenon across a large sample, quantitative methods like surveys or experiments would be more appropriate. Researchers must also consider practical factors such as time, budget, and access to participants, as these can influence the feasibility of certain methods. Ultimately, selecting the right data collection method ensures that the data is valid, reliable, and capable of addressing the research problem effectively.



Unit No 03: Sampling:

A sample in research is a smaller, manageable subset of a larger group or population that is selected for study. It represents the population, and data collected from the sample is used to make generalizations or draw conclusions about the entire population. The sample should reflect the characteristics of the population as closely as possible to ensure the research findings are valid and reliable.

Example: If you are studying the opinions of 1,000 students at a university about campus facilities, you might select a sample of 100 students to represent the whole population.

WHAT IS THE SAMPLING PROCESS?

The sampling process is the series of steps researchers follow to select the sample. Here's how it works in simple terms:

1. **Define the Population:** The population is the larger group that you want to learn about. For example, if you're studying student satisfaction in a university, your population might be all the students at that university.
2. **Determine the Sample Size:** Decide how many people or items you need in your sample. A larger sample usually provides more reliable results, but it also takes more time and resources.
3. **Choose a Sampling Method:** There are different ways to choose your sample. Two main types are:
 - **Probability Sampling** (Random Selection): Every member of the population has a known chance of being selected. Examples include simple random sampling, stratified sampling, or systematic sampling.
 - **Non-Probability Sampling** (Non-Random Selection): Some members of the population may not have a chance to be selected. Examples include convenience sampling or purposive sampling.

4. **Collect the Sample:** Using the chosen method, select your sample. For example, you might randomly choose 100 students from the entire student body.
5. **Analyze the Data:** Once you've collected the data from the sample, you analyze it to make conclusions about the larger population.

Example of Sampling Process: Imagine you want to study the eating habits of teenagers in a city. Instead of asking every teenager, you would follow the sampling process:

1. **Define the Population:** All teenagers in the city.
2. **Determine the Sample Size:** You decide to study 200 teenagers.
3. **Choose a Sampling Method:** You use **random sampling**, so every teenager in the city has an equal chance of being chosen.
4. **Collect the Sample:** You randomly select 200 teenagers and ask them about their eating habits.
5. **Analyze the Data:** You analyze their responses and make general conclusions about the eating habits of teenagers in the city.

By using this process, you can study a small group (sample) and still make conclusions that apply to the larger group (population).

DEFINE DIFFERENT KINDS OF SAMPLINGS:

1. Simple Random Sampling Simple random sampling is a method where every individual in the population has an equal chance of being selected. It's like drawing names out of a hat. Example: Imagine you have a class of 30 students, and you want to randomly select 5 students for a project. You write each student's name on a piece of paper, mix them up, and draw 5 names. Each student had the same chance of being chosen.

2. Systematic Sampling: Systematic sampling is a method where you select every n -th individual from the population, starting from a randomly chosen person.

Example: Suppose you want to choose 5 people from a list of 100. First, you randomly select one person from the first 5. Then, you select every 20th person after that. So, you might choose person 3, 23, 43, 63, and 83.

3. Cluster Sampling: Cluster sampling divides the population into groups (called clusters), then randomly selects entire clusters to sample. Each cluster is a mini-representation of the population. Example: If you are studying the eating habits of children in a city, you might divide the city into different neighborhoods (clusters). You randomly select a few neighborhoods and then survey all children in those neighborhoods.

4. Convenience Sampling: Convenience sampling involves selecting individuals who are easiest to reach or available, rather than randomly choosing them. It's the quickest and least expensive method but might not be the most accurate. Example: If you are conducting a survey about a new app, you may decide to ask people in your family or friends to participate because they are easily accessible.

5. Quota Sampling: Quota sampling is when you select people to match certain characteristics in proportion to their presence in the population. You stop when you reach a certain number for each group. If you're studying the opinions of men and women about a new product, and you want to have 50 men and 50 women in your sample, you continue selecting participants until you've reached those numbers.

6. Judgement Sampling: Judgement sampling is when the researcher selects individuals based on their judgment about who will provide the most useful information for the study. It's not random but based on the researcher's knowledge. Example: A researcher studying expert opinions on climate change might choose top environmental scientists or policy-makers as their sample, assuming they will provide the most valuable insights.

7. Snowball Sampling: Snowball sampling is a technique where existing study participants recommend other participants. It's useful for hard-to-reach or hidden populations. Example: If you're studying people who have rare health conditions, you

might start by interviewing one person, and then ask them to refer others with the same condition. This process keeps growing like a snowball, as each person helps find more participants.

Each of these sampling methods has its strengths and weaknesses, and the choice of method depends on the research objectives, population size, and available resources.

HOW TO SELECT APPROPRIATE SAMPLING TECHNIQUE?

Selecting an appropriate sampling technique is crucial for ensuring that the sample represents the population accurately and that the research findings are valid. The choice of sampling technique depends on various factors related to your research objectives, the population, the type of data needed, and available resources. Here are some key factors to consider:

1. Research Objectives

What is the aim of your research?

- If you're interested in generalizing results to the larger population, **probability sampling** methods (like **simple random sampling** or **systematic sampling**) are preferred because they allow you to make more reliable and valid inferences.
- If you need deep insights into specific individuals or small groups, **non-probability sampling** methods (like **judgment sampling** or **snowball sampling**) may be more suitable.

2. Nature of the Population

Is the population small or large?

- For **small populations**, you can usually use any sampling technique.
- For **large populations**, probability sampling methods, such as **simple random sampling**, might be more practical to ensure that every individual has a chance of being included without excessive resources.

Is the population homogeneous or diverse?

- If the population is **homogeneous** (similar), simple random sampling may be enough.
- If the population is **heterogeneous** (diverse), you might want to use **stratified sampling** (a form of probability sampling) to ensure that all important subgroups are represented.

3. Accuracy vs. Cost/Time

How important is accuracy?

- If **accuracy** and **representativeness** are critical, probability sampling techniques (e.g., **simple random sampling**, **systematic sampling**, or **cluster sampling**) are more appropriate.

How much time or money do you have?

- If you're on a **tight budget or time constraint**, **convenience sampling** or **quota sampling** might be quicker and cheaper, though these methods may introduce bias.

4. Data Type (Qualitative or Quantitative)

- **Quantitative Data:** If you're collecting numerical data or you need to make statistical comparisons, **probability sampling** is typically preferred. Techniques like **simple random sampling** or **systematic sampling** can be ideal.
- **Qualitative Data:** If you need to explore attitudes, experiences, or behaviors in-depth, non-probability sampling methods like **judgment sampling** or **snowball sampling** are often more suitable because they allow you to select individuals who are most likely to provide rich insights.

5. Population Access and Size

Is the population easy to access?

- If your target group is hard to access or hidden (e.g., drug users or experts), **snowball sampling** or **judgment sampling** might be more effective because they rely on referrals or the researcher's expertise.

How large is the population?

- If the population is very large, methods like **cluster sampling** can be effective, as you first divide the population into smaller groups (clusters) and then sample from those clusters.

6. Sample Size**How many people do you need to study?**

- For smaller sample sizes, non-probability methods (such as judgment sampling or convenience sampling) might be easier.
- For larger sample sizes, probability sampling techniques like simple random sampling or stratified sampling may be better to ensure that all relevant subgroups are included.

7. Research Constraints (Ethical and Practical Considerations)

- **Ethical Concerns:** Some techniques, like **snowball sampling**, may raise privacy or ethical concerns because they rely on participant referrals, which might affect the confidentiality of individuals.
- **Practical Constraints:** Consider logistical constraints, such as how easy it is to contact potential participants and whether they're willing to participate.
Convenience sampling can work well when access to participants is difficult.

Summary of Sampling Methods and When to Use Them:

1. **Simple Random Sampling:** Use when you want a random, unbiased sample, and when the population is large and accessible.
2. **Systematic Sampling:** Use when you have an ordered list and want a simple, efficient way of sampling at regular intervals.
3. **Cluster Sampling:** Use when the population is large and spread out, and you can divide it into clusters.
4. **Convenience Sampling:** Use when you need quick, inexpensive data, and the sample doesn't need to be perfectly representative.

5. **Quota Sampling:** Use when you need to ensure certain groups are represented in the sample, but don't have the time/resources for random sampling.
6. **Judgment Sampling:** Use when you rely on your knowledge or expertise to select the most appropriate individuals for the study.
7. **Snowball Sampling:** Use when studying hard-to-reach populations or those that are difficult to identify.

By considering these factors and understanding the strengths and weaknesses of each technique, you can select the most appropriate sampling method for your research.

ISSUES OF PRECISION AND CONFIDENCE IN DETERMINING A SAMPLE SIZE.

When determining the sample size for a study, precision and confidence are two critical factors that influence the quality and reliability of the results. These issues are important because they help ensure that the sample is large enough to reflect the population accurately, yet not so large that resources are wasted.

1. Precision: Precision refers to how close the sample estimates are to the true population values. A precise sample gives consistent results that closely match the actual population parameters.

- **Issue:** The **precision** of your sample depends on the variability in the population and the size of the sample. A smaller sample size can lead to **larger margins of error**, meaning that the results may be less accurate and less representative of the entire population.
- **Solution:** To achieve high precision, a larger sample size is often needed. However, this must be balanced against resource constraints like time and budget.

Example: Suppose you are conducting a survey to find the average income of employees in a company. A small sample might give you a number that's far off from

the actual average income of all employees, whereas a larger sample would provide a more precise estimate of the population's true average income.

2. Confidence: Confidence refers to how certain you are that the sample accurately represents the population. It's often expressed as a confidence level, such as 95% or 99%, which means that if the same study were repeated multiple times, the results would fall within a certain range.

- **Issue:** The **confidence level** reflects how much trust you place in the sample results. A higher confidence level (like 99%) typically requires a larger sample size. The confidence level represents the probability that the sample results fall within a certain margin of error.
- **Solution:** Increasing the sample size increases the **confidence** that the sample's results will reflect the true values of the population. However, higher confidence levels require more resources, so researchers need to find a balance.

Example: If you're conducting a poll about voter preferences and want to be 95% confident that your results are within 3 percentage points of the actual population proportion, you need a specific sample size. If you want 99% confidence, the sample size will need to be larger, which can be more expensive and time-consuming.

3. Balancing Precision and Confidence

- **Sample Size Formula:** The sample size is determined based on the desired level of **precision** and **confidence**. There are formulas used in statistics that calculate the sample size needed to achieve a certain level of confidence and precision.

The main factors that influence the required sample size are:

- **Confidence Level:** Higher confidence levels require a larger sample.
- **Margin of Error (Precision):** Smaller margins of error require a larger sample.
- **Population Size:** For large populations, sample size does not increase dramatically with a larger population size.

- **Population Variability:** More variability in the population requires a larger sample size to estimate the true population parameter accurately.

4. Practical Considerations:

- **Cost and Resources:** Determining an appropriate sample size involves balancing the desire for higher precision and confidence with available resources. Increasing the sample size increases the cost, time, and effort required for data collection and analysis.
- **Diminishing Returns:** After a certain point, increasing the sample size provides diminishing returns in terms of improved precision. This means that, beyond a certain sample size, the added accuracy is minimal compared to the increased cost.

Example of Calculating Sample Size: Imagine you want to estimate the proportion of people in a city who prefer a certain type of coffee. You want a **95% confidence level** and a **5% margin of error**. Based on the population size (e.g., 10,000 people) and estimated variability (e.g., 50% of people prefer this coffee), a sample size calculator or formula will tell you the minimum sample size required to achieve those precision and confidence goals.

- **For a 95% confidence level and 5% margin of error**, the sample size might be around **384** for a population of 10,000 people.
- If you want **99% confidence** or a **3% margin of error**, the required sample size would increase, possibly to **1,000 or more**.

Conclusion:

The key issues in determining sample size are ensuring the right **precision** (how close the sample estimate is to the true population value) and the right **confidence level** (how sure you are that your sample represents the population accurately). A larger sample size generally increases both precision and confidence, but it requires

more resources. Therefore, researchers must balance these factors carefully to make the most effective and practical choice for their study.

